

GE Healthcare

MR - 1.5T Signa® HDx & 3.0T Signa® HDx Release Notes and Errata

1.5T Signa® HDx & 3.0T Signa® HDx
Operator Manual, English
5159410-100
Revision 2
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Chapter 1

Release Notes

Purpose and Scope

This Release Note & Errata covers the 1.5T Signa® HDx & 3.0T Signa® HDx product. The document is divided into two chapters.

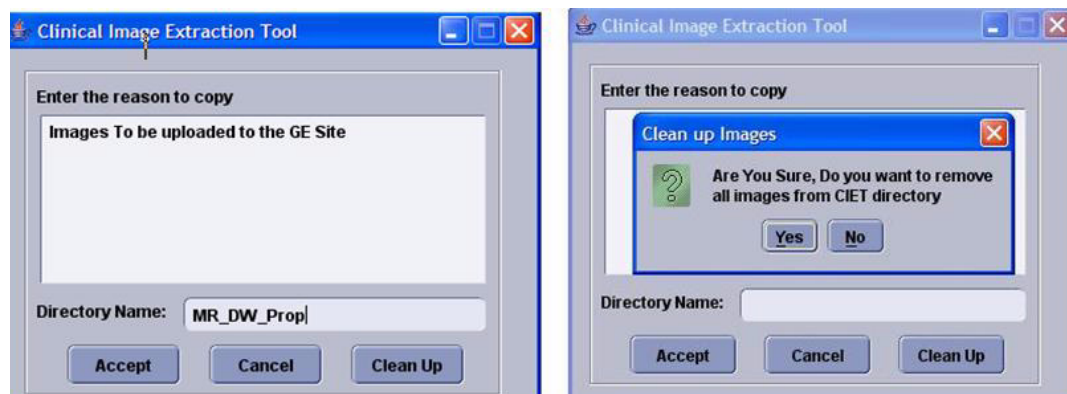
- Chapter One provides the release notes that include additional information not available in the EXCITE HDx Online Help.
- Chapter Two identifies errata information.

Clinical Image Extraction Tool (CIET)

CIET, a new productivity tool, allows InSite to pull images for analysis from your site. A new feature of CIET (the new productivity tool that allows InSite to pull images for analysis from your site), contains a new option called Clean Up.

Clean Up clears out the directory of images saved in CIET. It is important to keep the CIET directory as small as possible because it shares the same space as your Exam/Series/Image directory.

Click **CleanUp> OK** to empty the CIET directory.



Multi-Coil Spectroscopy

MCSI (multi-channel CSI) can now be used with asset compatible multi-channel coils. Prescribe the MCSI scan by selecting 2D or 3D imaging mode and Probe P pulse sequence. The coil type, coil and configuration is listed below. Note that an Asset calibration scan is required.

Coil Type	Coil	Configuration
Head	Invivo/MRI Devices	8HR Brain
Head	Invivo/MRI Devices	HDx 8HR Brain
Head/Neck	HD NV Array by USAI	HDNV Array_A
Head	HD NV Array by USAI	HDNV Head_A
Head/Neck	HD NV Array by USAI	HDNV Array
Head	HD NV Array by USAI	HDNV Head
Head/Neck/Spine	HNS by USAI	HNS Head
Head/Neck/Spine	HNS by USAI	HNS NV Full
Head/Neck/Spine	HNS by USAI	HNS NV Head

LAVA XV

LAVA XV is now compatible with the oblique plane.

MR Echo

When prescribing Scan&Save slice groups, slices can be added or removed from the group, without changing the group angle, by holding CTRL key while adding/removing slices.

CartiGram (previously identified as T2 Mapping)

T2 Mapping is now called CartiGram but is still labeled as T2 Mapping in the online operator's manual.

CartiGram is compatible with both quadrature and phased-array coils (see below).

Field	Coil Type	Coil	Configuration
1.5 T	Extremity	Invivo/MRI Devices	PE (Precision Eight) Wrist
1.5 T	Extremity	Invivo/MRI Devices	HD Quad Knee
1.5 T	Extremity	Invivo/MRI Devices	HD TRknee PA
3.0 T	Extremity	Invivo/MRI Devices	HD Quad Knee
3.0 T	Extremity	Invivo/MRI Devices	HD TRknee PA

Multi-Phase

Delay after Acquisition: When selecting a delay after acquisition, note that if Variable Delays is selected, the value is rounded to 1 second and the range is 1 to 300 seconds. If Variable Delays is not selected, the value is rounded to 0.001 second and the range is Minimum to 300 seconds.

Auto Subtract: When performing an Auto Subtract, you can select which series to subtract (First Phase of Same Series, Previous Series or Series Number) unless you have Mask Phase selected. If Mask Phase is selected, Auto Subtract is automatically performed against the mask images.

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Chapter 2

Errata

Purpose and Scope

This chapter identifies known problems or issues in this release of the system. This chapter is divided into sections based on applications. In addition, this chapter provides descriptions of the situation in “What Happens” and then the “Work Around” to resolve the problem.

Scan

What Happens: The following scenario may happen with the first series of a scan prescription.

1. Select the protocol and have the first series in an InRx state and this first series has an offset, for example I950.
2. Go into the scan room, position the patient and set the landmark, for example at the patella.
3. Press the **Move to Scan** button on the Magnet Control panel and the system advances to the offset in the series and not the landmarked location.

Work Around: To have the system advance to the landmark in the above scenario and not the offset, do not select the protocol until after you have pressed **Move to Scan** button on the Magnet Control panel.

What Happens: The number of radial slices on the Graphic Rx window is greater than one. You left-click to deposit a group of slices, which results in a radial prescription. You click **Erase All** and left-click to deposit a non-radial prescription. Another radial prescription is deposited.

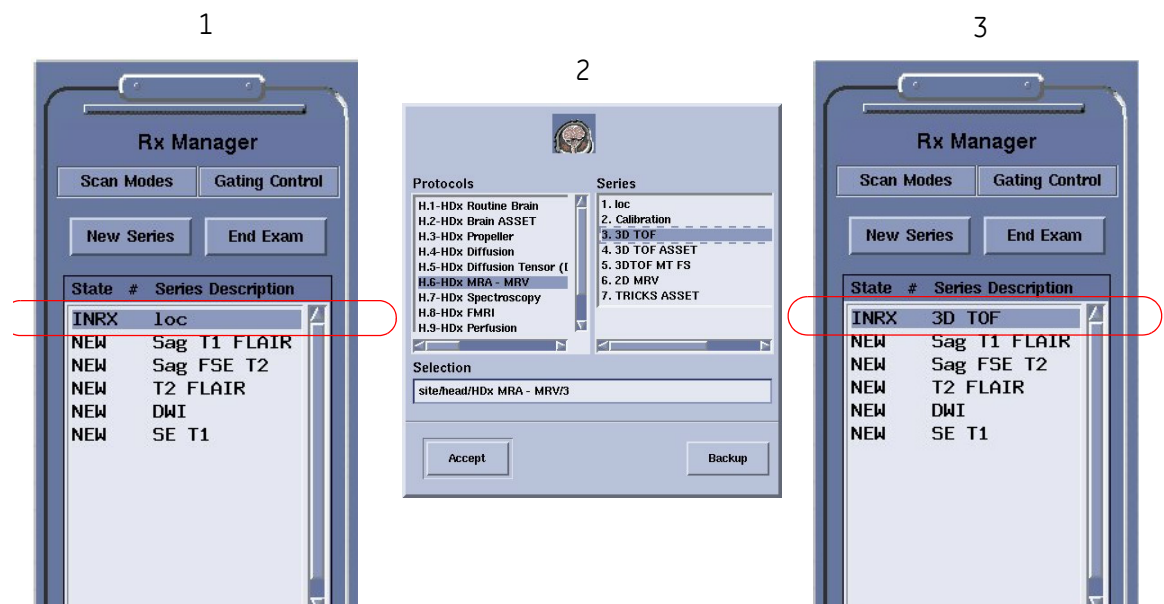
Work Around: You must set the number of radial slices to 1 so that when you deposit a set of slices it is not a radial prescription. Clicking **Erase All** does not reset the number of radial slices to 1.

What Happens: The Sat bands are shifted right to left on a multi-slice multi-angle oblique axial image that is prescribed from a coronal image.

Work Around: Since Sat bands are orthogonal and prescribed on a straight coronal, the final images will have the SAT Bands shifted to one side. This is to be expected, position SAT bands on all orthogonal or all oblique images to maintain proper SAT band placement.

What Happens:

1. The first series in the Rx Manager is highlighted (1).
2. Click the **Protocol Man** icon and select a protocol to add to the Rx Manager. Click **Accept** from the Patient Protocol list (2).
3. The highlighted series in the Rx Manager is over written with the first series of the protocol selected in step 2 (3). The remaining series are added to the end of the downloaded protocol.



Work Around: Always select **New Series** when adding a protocol to the Rx Manager to prevent over writing a series in the Rx Manager.

What Happens: When a TPS Reset is started and the table is moved before the TPS Reset is completed, the landmark is lost.

Work Around: To retain the landmark, do not move the table until the TPS Reset is completed.

What Happens: Real Time is running an FGRE acquisition. You define a Start and End location from the Review tab. You click **[Save Range]** and the Save Range pop-up displays. While it is saving, you switch desktops. When you return to the Scan desktop the display message box is opened and blank, therefore it cannot be closed.

Work Around: Do not switch desktops until the save range message box disappears.

What Happens: A misleading message is displayed when Auto Start is active and Advance to Scan is selected. The message in the Scan Operations area states: *Prescanning*. The system does not start the prescan until the scan room door is shut. The message should read: *Downloaded*.

Work Around: None

What Happens: When booting the system up a message appears instructing you to make a monitor resolution selection.

Work Around: Ignore the message and wait for it to disappear. Wait until all messages are removed from the screen and the Scan desktop is complete before clicking any desktop icon.

What Happens: Image blurring on 3T extremity images may occur with an off-center FOV and the Zoom gradient coil.

Work Around: Use one of the following methods to eliminate this problem:

- Use Phase Correction with Zoom mode for off-center FOV imaging.
- Use Whole mode instead of Zoom mode for off-center FOV imaging.

What Happens: When using 2D FSE with VERSE, NPW, and an odd NEX or an even NEX that is not a multiple of four, a fine line artifact can result.

Work Around: Use four, eight, or twelve (multiples of 4) NEX with NPW.

- Turn off NPW and use two, four, six, etc. (even) NEX, and either increase the FOV or swap phase/frequency direction to avoid phase wrapping.

What Happens: The system has a limitation of 10,000 non-fMRI images per series. The system will allow you to prescribe more than 10,000 images within a series, download and prescan the series. When prescan is completed, an error message is posted stating that scan has stopped and please try again.

Work Around: Calculate the total number of images to be acquired within a series prior to saving the series and downloading it. If the number exceeds 10,000 for a non-fMRI scan, adjust the scan parameters accordingly (reduce the number of phases or scan locations).

Coils

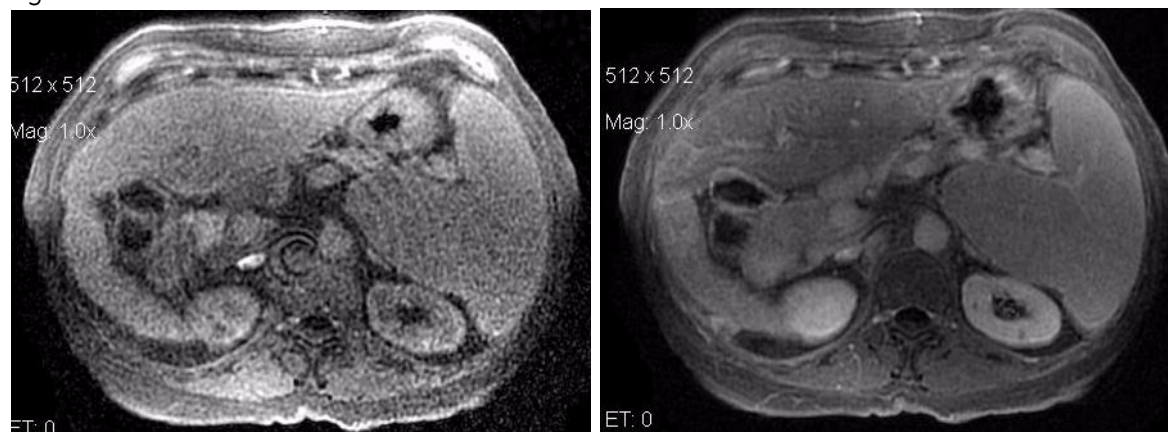
What Happens: When prescribing a cal scan for the HDNC MEDRAD coil, the default FOV is 48 cm. Using this FOV can result in image quality problems, particularly if PURE is used.

Work Around: Change the FOV to 30 cm.

What Happens: Frequently, when using the 8-channel Body Array coil, Full FOV with LAVA, a bull's eye artifact results.

Work Around: Use either upper or lower FOV for the 8-channel Body Array.

Figure 2-1 8-channel body array, full FOV on left, 8-channel body array upper FOV on right



Applications

MR-Echo

What Happens: You are scanning an MR-Echo Functions, Time Course or Myocardial Evaluation acquisition and the scan stops and a scan failure error occurs. You click **[Restart]** on the MR-Echo real time window and the button becomes non-selectable and the scan does not restart.

Work Around: Click **[Exit]** from MR-Echo real time window, which displays the MR-Echo prescription window. Click **[Scan]** to resume the scan.

What Happens: The following scenario can occur in MR-Echo.

1. An MR-Echo is prescribed and the MR-Echo Realtime screen is displayed.
2. While realtime images are being acquired, you click **Scan and Save** icon.
3. No slices are prescribed in any of the viewports for 30 seconds. After 30 seconds, Realtime screen times out and the viewports become blank or black.
4. Click **[Restart]** to restart Realtime and then you can click **Scan and Save** and deposit slices.
5. If instead of completing step 4, you move the cursor into the black viewport and attempt to **Scan and Save** the system allows you to deposit slice lines on a blank viewport. Clicking **[Scan]** at this point removes the graphic objects and you are no longer able to **Scan and Save** images.

Work Around: If you complete step 5, you must copy/paste the series and start again.

What Happens: In MR Echo, if you try to change a landmark after the series is downloaded, a message is posted advising you that the landmark is no longer valid.

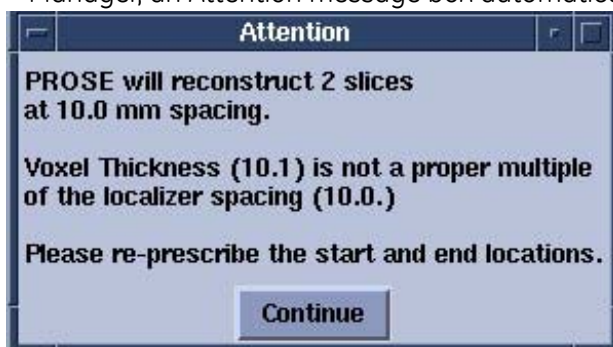
Work Around: Start a new series.

PSD

What Happens: PROBE-P spectroscopy screen saved series are not PPS (Performed Procedure Step) compliant.

Work Around: None

What Happens: When you View/Edit a 3D CSI PROSE or PROBE series in the Rx Manager, an Attention message box automatically displays.



Work Around: Click **Continue** to close the message box. Re-prescribe so the voxel thickness is equal to the localizer spacing. In this example, localizer spacing is 10, so voxel thickness should be 10. This message box also displays after you click Save Series for a 3D CSI PROSE or PROBE series.

What Happens: STIR breast images may have a dark shadow over one of the breasts giving the appearance of a dielectric effect.

Work Around: Use a sagittal T2 FSE fat SAT acquisition or an axial/sagittal long TE, T2 FSE acquisition.

What Happens: You are prescribing a Spiral scan. If you delete the SSRF option from the Imaging Option field on the Imaging Parameters area of the scan desktop rather than from deselecting it on the Imaging Parameters pop-up window, then the prescription user interface greys out (becomes invalid).

Work Around: Do not delete SSRF from the Imaging Options text field.

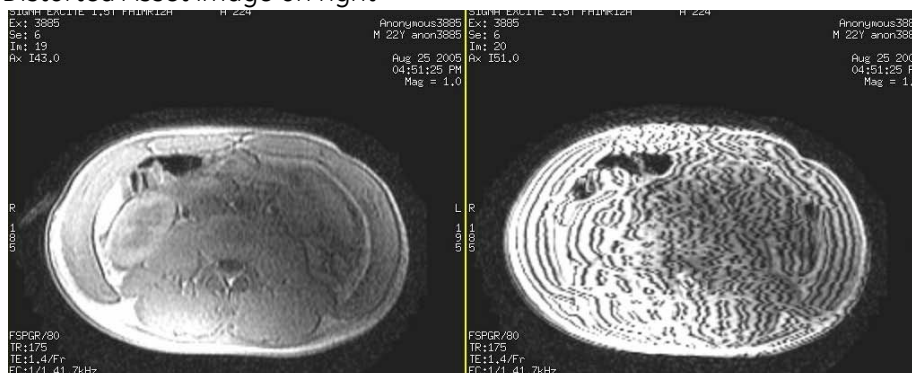


Only delete an Imaging Option by deselecting the radio button from the Imaging Parameters pop-up window.

Imaging Options

What Happens: Asset images may appear distorted when a too thin slice thickness is used in the cal scan.

Figure 2-2 Distorted Asset image on right

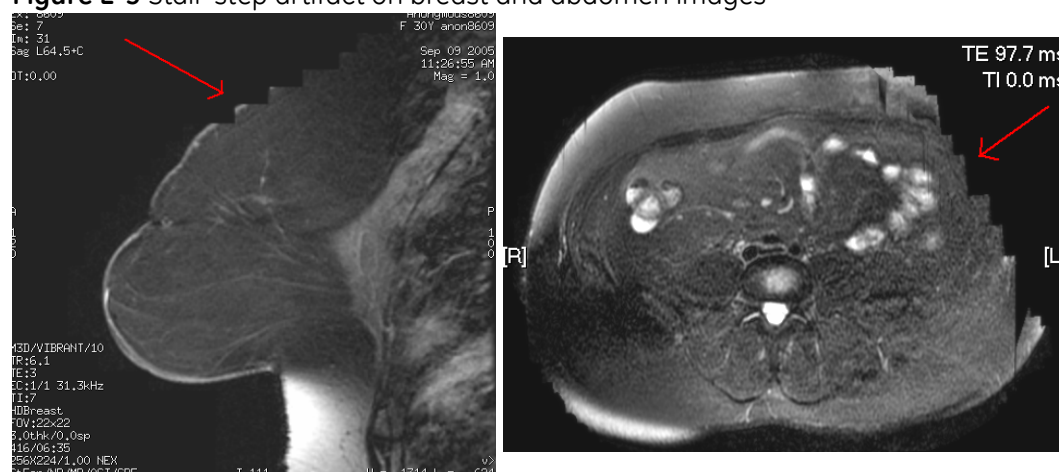


Work Around: Use a 20mm slice thickness cal instead of a 9mm slice thickness.

What Happens: When a stair-step type of artifact appears on the edge of the anatomy, it is due to one of the following:

- A channel(s) is out on the coil.
- The calibration extends too far beyond the coil.

Figure 2-3 Stair-step artifact on breast and abdomen images



Work Around: Have your service engineer check your coil for bad channels. Do not acquire data beyond the calibration range. If you need to image this area, repeat the cal file and extend the scan range.

What Happens: When the Imaging Options of ASSET and Respiratory Trigger are selected with a 2D abdominal scan, mis-registration occurs due to breathing motion.

Work Around: Do not select the Respiratory Trigger Imaging Option with a 2D abdominal scan that has ASSET turned on.

What Happens: The Scan button on the keyboard is illuminated for an fMRI scan. When pressed, it will NOT start the scan.

Work Around: Only use the **[Scan]** button on the Scan Operations area of the Scan desktop to activate the functional scan.

Protocols

Protocol Exchange

What Happens: Use Protocol Exchange to export an MR-Echo series. Only the RealTime parameters are exported and all the scan parameter values for each of the other MR-Echo tabs are not exported.

Work Around: Manually record the scan parameter values for each of the MR-Echo tabs.

What Happens: With Protocol Exchange, if your protocols have non-English characters in the protocol description, these characters are incorrectly displayed on the Protocol Exchange window.

Work Around: Do not use non-English characters.

What Happens: If you are importing a protocol using Protocol Exchange and the protocol you are importing is already on the system, an over-write prompt appears. If you enter a non-English character text field, the protocol will appear in the Site protocol library but no series will be attached to the protocol.

Work Around: Do not use non-English characters when entering a protocol name in the overwrite message text box.

What Happens: When using a USB port device (smart stick or jump stick) to export protocols from the Protocol Exchange feature, the USB device must be in a read mode. If it is not in a read mode, when you export the protocols, it appears that the protocols have been saved, i.e. there are no error messages. When you view the series within each protocol from the USB device you will not see any protocols.

Work Around: Check the properties of the USB port device and make sure it is in a read state before you plug it into the MR computer.

Archive/Network

What Happens: A DICOM Structured Report series generated from FuncTool or a Saved State series generated from the Viewer cannot be restored from an EXCITE HD MOD to a 9.1, 10.0, or 11.0 system. Nor can these series be networked from an EXCITE HD system to a 9.1, 10.0, or 11.0 system. When an EXCITE HD MOD is placed in a 9.1, 10.0, or 11.0 system, the MOD becomes a read only MOD. In other words, you cannot store 9.1, 10.0, or 11.0 images onto an EXCITE HD MOD.

Work Around: None.

What Happens: Networking a 12.0 exam including SR or /GSPS object series to a 9.1 or 11.0 system is successful and no errors are reported. However, pushing or pulling a 12.0 SR or /GSPS object series from a 9.1/11.0 system will cause the exam to be paused on network queue with error to operator 'Receive failed.'

Work Around: Don't network 12.0 GSPS or SR series to 9.1 or 11.0 systems.

What Happens: Auto network by series should transfer the images in the series once scanning has ended and the images are in the Browser. If an MR-Echo scan has been bookmarked and you immediately exit the series, the images may not transfer.

Work Around: If, by the time you have ended the exam and the series has not yet been networked to the end destination, manually transfer the series by completing the following steps:

1. From the Browser list, click the desired series to highlight it.
2. Click **[Network]**.
3. Click **[Send Series]**.

Display

What Happens: The image annotation for EPI images is FE (an abbreviation for Fractional Echo) for min TE and EF for min Full.

Work Around: None.

What Happens: Images post processed with ClariView that have SCIC applied, can have large portions of the image that are saturated with maximum pixel value (appearance of bright white).

Work Around: Only apply SCIC from the Image Enhance Additional Parameters selection during the scan prescription process.

What Happens: In IVI, there are Navigator protocols on the Additional Guides window. You can select these protocols and the Navigator wizard appears in the lower left viewport, but you cannot create a navigator view. Navigator is not available on MR systems.



Work Around: None. Navigator is available on AW systems.

FuncTool

What Happens: DWI images cannot be used as a spectroscopy localizer in FuncTool. Note that you can prescribe a spectroscopy scan from a DWI image, but FuncTool will not display it as the localizer.

Work Around: From the scan desktop, use a non-DWI image as the localizer for placing a spectroscopy VOI.

What Happens: Data Export current button doesn't update automatically

1. **[Current]** is selected on Image Range Selection.
2. Save the current image and then page through the images.
3. Save another image.
 - ♦ The system does not automatically update to that image - instead, it resaves the previously saved image.

Work Around:

1. Click **[All]** from Image Range Selection.
2. Reselect **[Current]** and it will update.

What Happens: When in FuncTool and you **ScreenSave + Add to Report** and then view the image from the Generate Report window, the image displayed is not the image you saved. A test image is saved and displayed.

Work Around: None.

What Happens: The Structured Report SR Viewer allows you to display a report that was generated in FuncTool. When you select the series from the Browser to be viewed and then click SR Viewer, the images cannot be viewed.

Work Around: None.

What Happens: If you load a PROSE CSI data set acquired with a multi-channel coil into FuncTool, the system automatically launches the single voxel Breast protocol.

Work Around: From the FuncTool desktop, click [New Protocol] and select the desired protocol button.

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